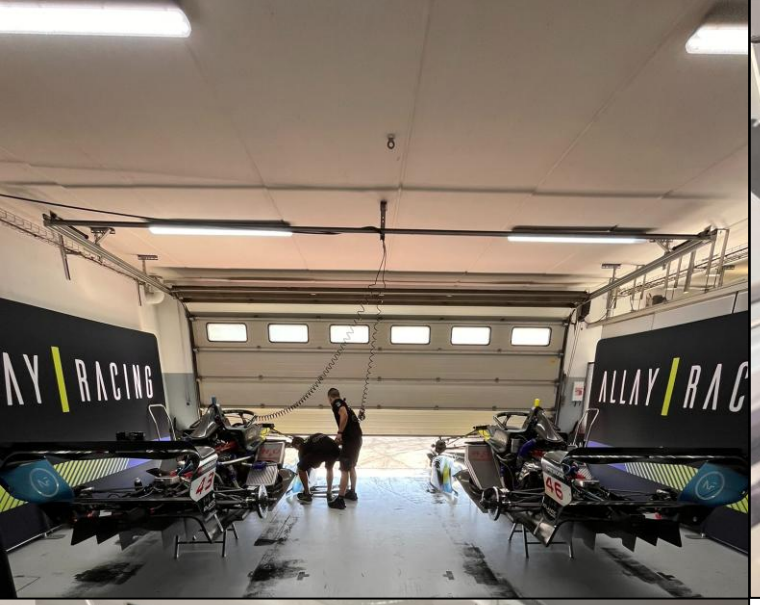


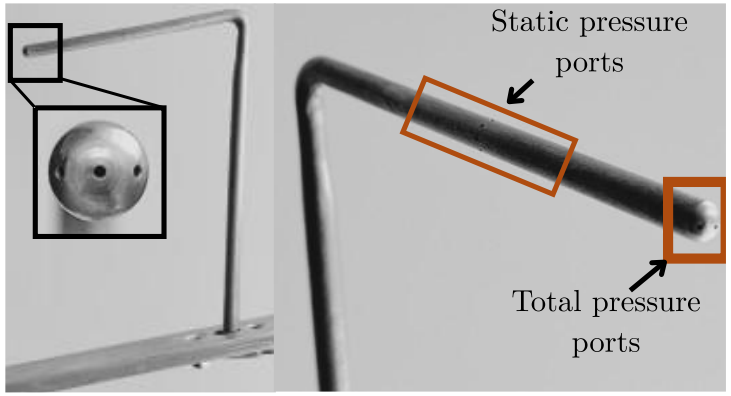
Aerodynamics Portfolio

Adrian Ceballos

MSc Dissertation Project:
Yaw Probe Calibration with Wind Tunnel Testing

Objectives:

1. Assess the new Oxford Brookes wind tunnel (WT) and verify the adequate operation multiple data acquisition systems.
2. Map the pressure data at multiple angles of attack (AoA).
3. Characterise the response under dynamic yaw motions coding a MATLAB algorithm.
4. Document the calibration guidelines based on previous results.



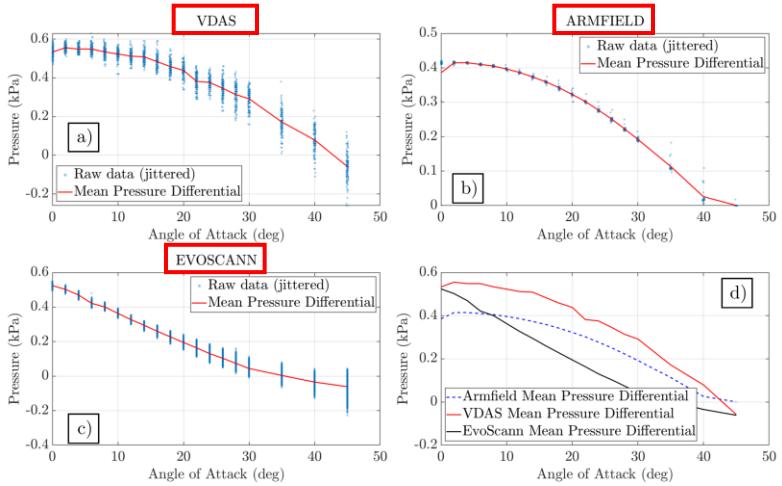
1. Assessment of the Equipment and Data Acquisition Systems

Troubleshooting control faults between the WT software and the PC

- Installed and configured the WT on my laptop
- Conducted preliminary tests with a NACA wing profile

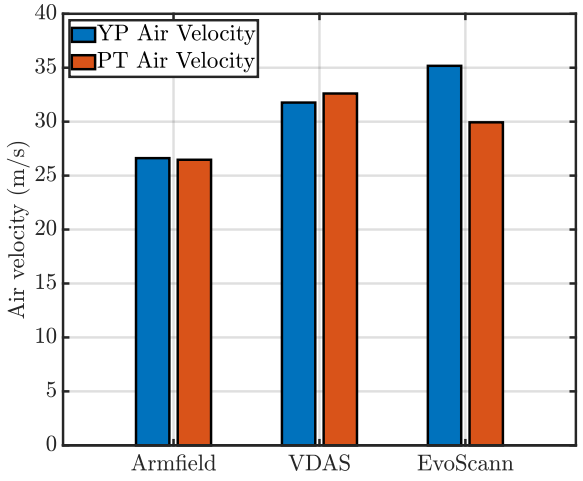
Evaluation of pressure measurements of 3 data acquisition systems

Compared the characteristic response in the mean pressure differential

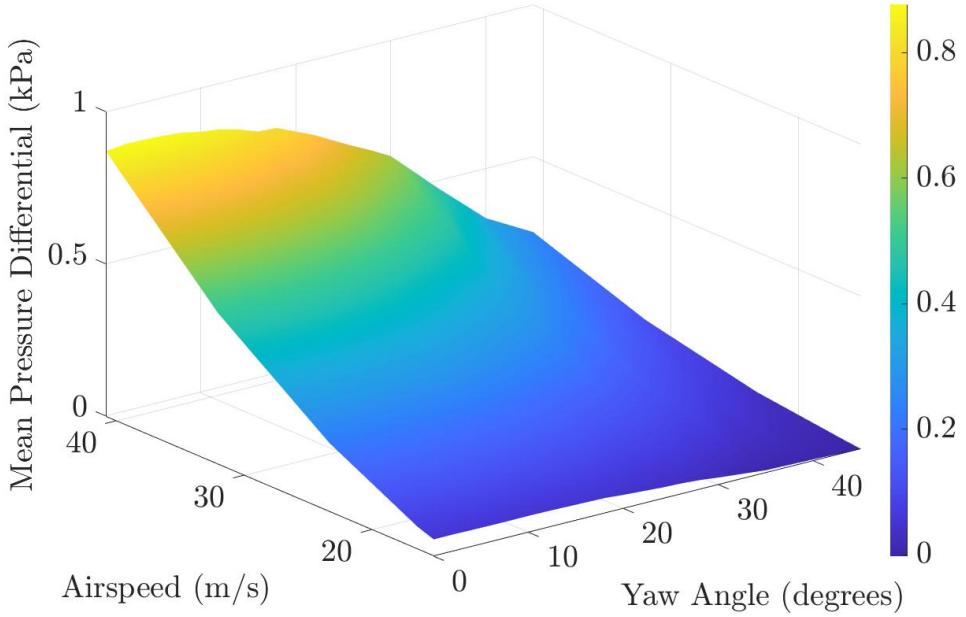


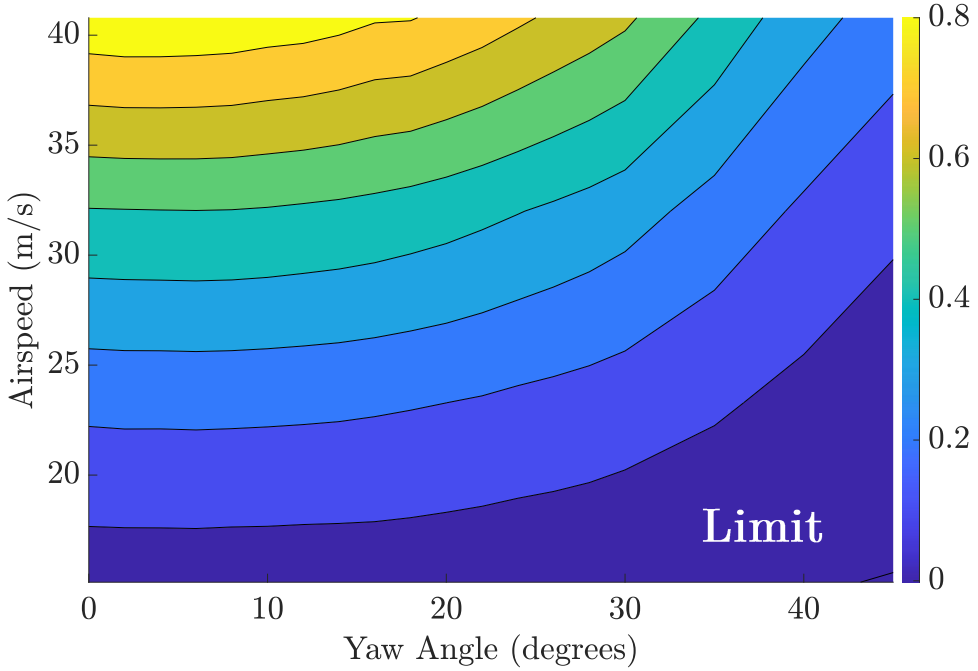
Calculation of air velocity using a Pitot Tube and a Yaw Probe

Acquired data with the 3 previous systems and calculated the air velocity



2. Mapping the Pressure Data at Multiple AoA



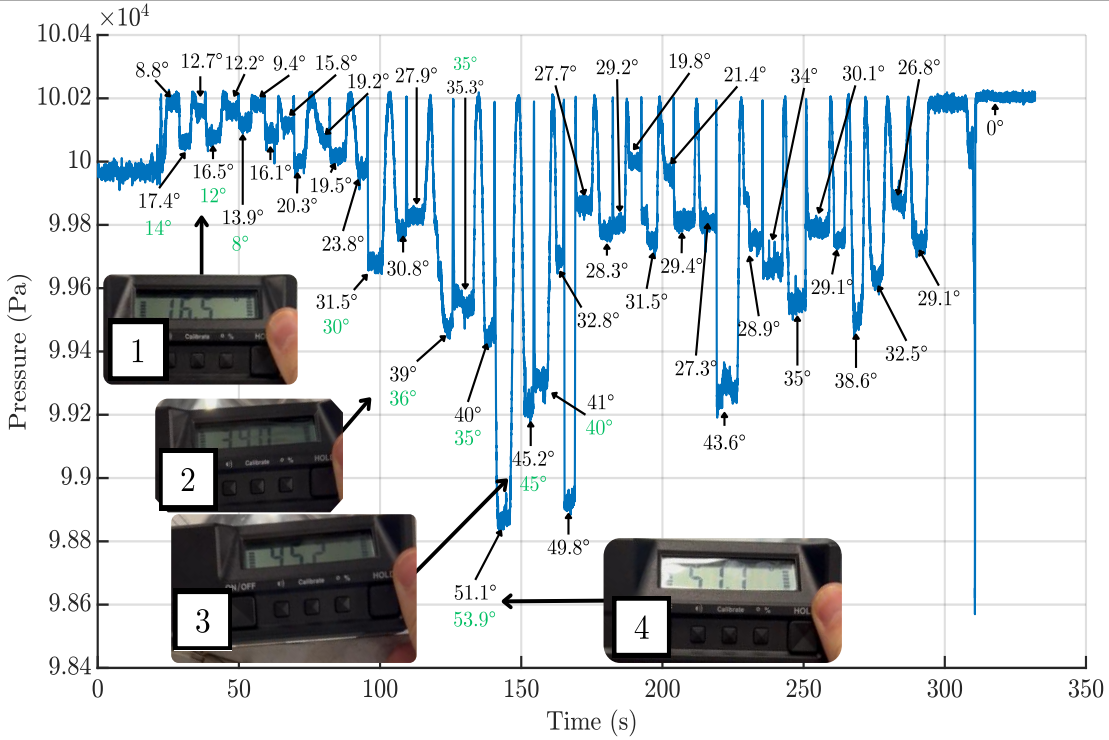


- Mapped the yaw probe by conducting wind tunnel tests at 6 airspeeds ranging from 40 to 6 m/s and AoA from 0 to 45°.
- Found the operational limit of the instrument at low airspeeds and AoA.
- Coded a MATLAB tool to process high-resolution EvoScann data and to generate the maps above. This was infeasible with Excel.

3. Yaw Probe Characterisation with Dynamic Testing

Set-up used during the dynamic testing stage, using:

- FISCO inclinometer.
- EvoScann pressure scanner.
- Yaw probe.
- Armfield C30 wind tunnel.

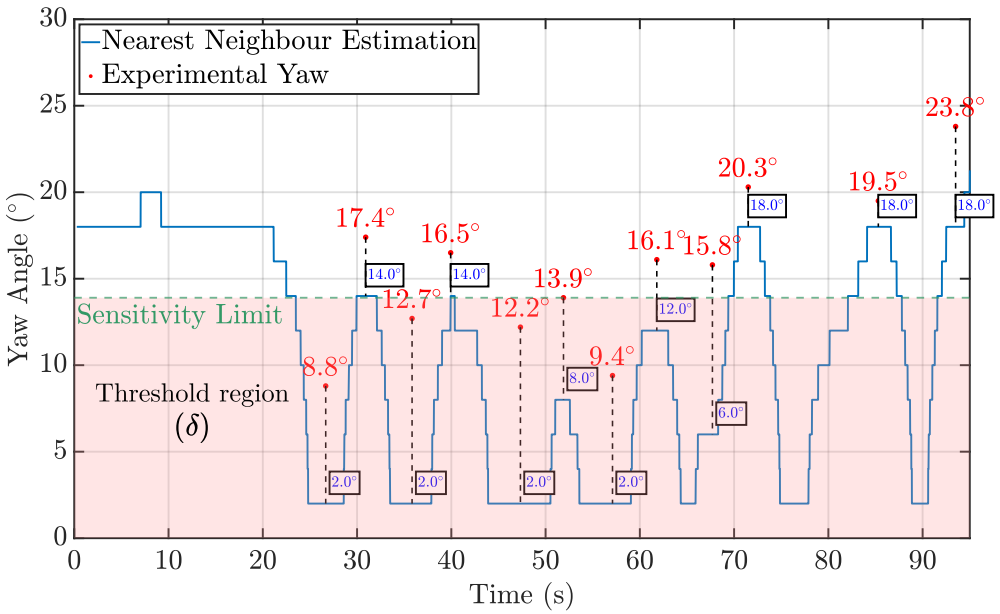


Total pressure trace of the dynamic yaw motions.

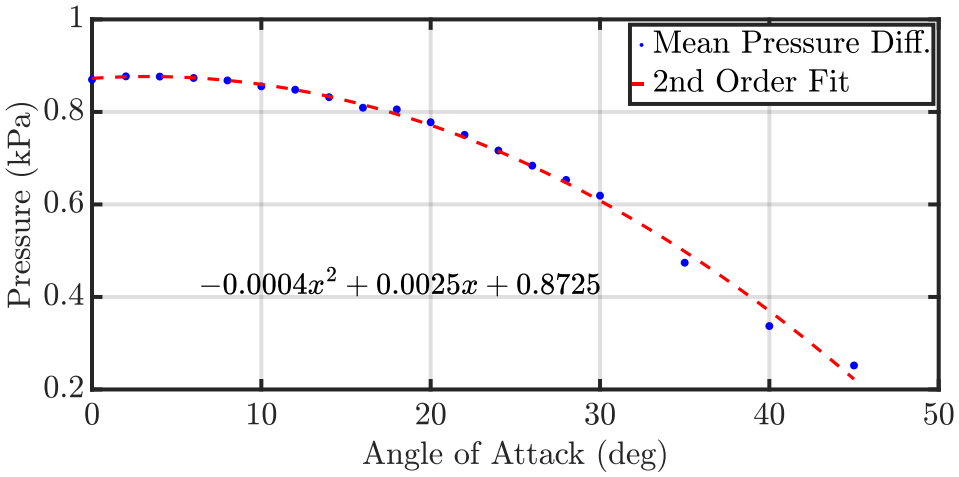
- The black numbers are the measured readings from the inclinometer.
- The green numbers are the estimations from the MATLAB algorithm that uses:
  - Nearest neighbour interpolation.
  - Piecewise Cubic Hermite Interpolating Polynomial (PCHIP) extrapolation.

4. Calibration Guidelines

- Sensitivity threshold region defined after analysing the pressure trends obtained from the dynamic testing stage.
- Yaw probe calibration curve and characteristic equation derived from the results in the mapping process.



Definition of the sensitivity threshold region.



Yaw probe calibration curve.